



FIRST SEMESTER 2026-2027
COURSE HANDOUT

Date: 25.07.2026

In addition to Part I (General Handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

Course No : CHE F213
Course Title : Chemical Engineering Thermodynamics
Instructor-in-Charge : Prof. Mohit Garg
Instructor(s) :
Tutorial Instructors : Prof. Pradipta Chattopadhyay and Prof. Aditya Singh

1. Course Description:

Applications of first and second laws; relations between state properties; chemical equilibria in reacting and non-reacting systems; statistical concepts, and brief exposure to irreversible thermodynamics; extensive problem assignments throughout. [Review of work, heat, reversible and irreversible processes, First Law applications to closed and open systems, Second Law, Entropy, Heat effects, Equations of state and generalized correlations for PVT behavior, Maxwell relations and fluid properties estimation; Residual and excess properties, Partial molar quantities; Gibbs-Duhem Equation, Fugacity and Activity Coefficient models, Vapor-Liquid equilibria, chemical reaction equilibrium.

2. Scope and Objective of the Course:

- Understanding laws of thermodynamics and applying the concept for heat, work and energy calculations.
- Analyze essential conditions (temperature, pressure, composition, etc.) for simple physical and chemical changes.
- Thermodynamic properties calculation for ideal and real fluids.
- Apply thermodynamic principles in estimation/calculation of thermodynamic efficiency
- Application of thermodynamic principals for understanding the criterion (ΔG , equilibria constant, etc.) of simple chemical reaction at equilibria, degree of conversion, composition and effect of P, inert gas addition, excess reactants, products on conversion, etc.
- VLE calculations for phase-equilibria.

3. Students learning outcomes:

1. Develop ability to analyze heat, energy, and work requirement/transfer for simple physical and chemical processes using laws of thermodynamics.
2. Applying principles of thermodynamics to chemical engineering systems.
3. Industrial design of reactors, separators and utilities using principles of thermodynamics.
4. Thermodynamic efficiency calculation for chemical engineering processes.



5. Feasibility of chemical engineering processes based on thermodynamic properties calculations.

4. Text Books:

T1: J. M. Smith, H C Van Ness, M. M. Abbott and M. T. Swihart, *Introduction to Chemical Engineering Thermodynamics (8th ed.)*, Tata McGraw Hill, special Indian Edition 2024.

Reference Books:

- J. Richard Elliot, Carl T. Lira, *Introductory Chemical Engineering Thermodynamics*, 2nd Edition, Pearson, 2015
- Koretsky Milo D., *Engineering and Chemical Thermodynamics*, Wiley Student Edition, 2004.
- Sandler, Stanley I., *Chemical, Biochemical and Engineering Thermodynamics*, 4th Edition, Wiley (India Pvt. Ltd.: Wiley Student Edition), 2006.

5. Method of conduct of the course:

Classroom teaching & Tutorial classes

6. Course Plan:

Module: Lecture No.	Topics to be Covered in Lecture (L) & Tutorial (T) Sessions	Reference Ch./Sec. #	Learning Outcome
M1: 1 Introduction	L1: Scope and Objectives of course, Course structure, work, heat, energy, temperature.	T1: 1.1- 1.9	Understand 1. Scope and objectives, and structure of course 2. Basic concepts: work, heat, energy, temperature.
M2: 2 - 3 First Law of Thermodynamics & Basic Concepts	L2: First law, Energy balance for closed system, Thermodynamic state and state functions, Equilibrium, Phase rule, Reversible process, Const-V and Const-P processes, Enthalpy, Heat capacity.	T1: 2.1-2.8	1. Identify different forms of energies: work, heat, internal energy, heat capacity, enthalpy, etc. 2. Understand applications of 1 st law 3. Use state functions for energy calculations of simple practical thermodynamic problems
	L3: Mass and energy balance for open systems.	T1: 2.9	
	T1: Exercise problems on 1 st law & basics of thermodynamics.	T1: Ch1 & Ch2 Problems	
M3: 4 – 7 Volumetric	L4: PVT behavior of pure substances, Virial	T1: 3.1-3.2	1. Calculate/analyze thermodynamic state



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Properties of Pure Fluids	equations.		parameters/equations (P, V, U, C, H, etc.) for pure fluids at particular conditions (P, V, T, etc.) 2. Calculate/analyze work & energy requirement/transfer for pure fluids in simple industrial processes.
	L5: Ideal gas, Applications of Virial equations.	T1: 3.3-3.4	
	L6: Applications of Virial equations, Cubic equations of state.	T1: 3.4-3.6	
	L7: Generalized correlations for gases and liquids.	T1: 3.7-3.8	
	T2: Exercise problems on Volumetric Properties of Pure Fluids.	T1: Ch3 Problems	
M4: 8 – 9 Heat Effects	L8: Sensible heat effects, Latent heat of pure substances Standard heats of reaction, formation and combustion, Temperature dependence of ΔH .	T1: 4.1-4.6	1. Calculate/analyze the requirement/transfer of heat energies and related parameters of fluids applicable in simple chemical/physical processes 2. Utilize heat effects of reactions for defining operation conditions of industrial processes
	L9: Heat effects of industrial reactions.	T1: 4.7	
	T3: Exercise problems on Heat Effects.	T1: Ch3 Problems	
M5: 10 – 12 Second Law, Entropy and Third Law	L10: Statements of second law, Heat engines, Thermodynamic temperature scale, Entropy, Entropy change for an ideal gas, Mathematical statement of the second law.	T1: 5.1-5.5	1. Understand applications of 2 nd law, importance of idea of entropy. 2. Calculate entropy change and different works (ideal, actual, and lost) and related properties for the simple process related changes.
	L11: Entropy balance for open Systems, calculation of ideal work.	T1: 5.6-5.7	
	L12: Lost work, Third law of thermodynamics, Entropy from the microscopic view point.	T1: 5.8-5.10	
	T4-T5: Exercise problems on Second Law, Entropy and Third Law.	T1: Ch3 & Ch4 Problems	
M6: 13 – 17	L13: Property (H, U, A,	T1: 6.1	1. Estimate/calculate heat



Thermodynamic Properties of Fluids	G, S, P, V, T, etc.) relations for homogeneous phases.		and work quantities for the simple and homogeneous fluids in industrial process changes. 2. Estimate/calculate actual thermodynamic properties (P, V, G, A, H, S, etc.,) for the simple and homogeneous fluids in industrial processes.
	L14: Residual properties and Residual properties calculation by equations of state.	T1: 6.2-6.3	
	L15: Residual properties calculation by equations of state.	T1: 6.3	
	L16: Two-phase systems, Thermodynamic diagrams and tables.	T1: 6.5-6.7	
	L17: Generalized property correlations for gases.	T1: 6.4	
	T6-T7: Exercise problems on Thermodynamic Properties of Fluids.	T1: Ch6 Problems	
M7: 18 - 19 Applications of Thermodynamics to Flow Processes	L18: Duct flow of compressible fluid.	T1: 7.1	1. Estimate/calculate Q, W, H, S, G, etc., for the flowing fluids in simple industrial process structure and equipment. This chapter will help students to understand fluid flow and related processes addressed in Fluid Mechanics (CHE F212) course.
	L19: Duct flow of compressible fluid, Turbines,	T1: 7.1-7.2	
	L20: Turbines, Compression processes.	T1: 7.1-7.3	
	T8-T9: Exercise problems on Applications of Thermodynamics to Flow Processes	T1: Ch7 Problems	
M8: 20 – 24 Introduction to Vapour/Liquid Equilibrium	L21: Nature of equilibrium, Phase rule; Duhem's theorem.	T1: 12.1-12.2	Apply thermodynamic principles in estimation/calculation of T, P, phase compositions (x & y), of simple VLE equilibrium systems and changes. This chapter will help students to understand mass transfer related (distillation, adsorption, extraction, drying, etc.,) simple industrial techniques addressed in Separation
	L22: VLE: Qualitative behaviour, Simple models for VLE.	T1: 13.3	
	L23: VLE by Modified Raoult's law.	T1: 13.3	
	L24: VLE from K-value correlations.	T1: 13.4	
	T10-T11: Exercise problems on Vapour/Liquid Equilibrium.	T1: Ch12 & 13 Problems	



			Processes I (CHE F244) and Separation Processes II (CHE F313) courses.
M9: 25 - 29 Basic Concepts of Solution Thermodynamics: Theory	L25: Fundamental property relation, Chemical potential and Phase equilibrium, Partial properties.	T1: 10.1-10.3	Apply thermodynamic principles in estimation/calculation of enthalpy, entropy, free energy, temperature, pressure, volume, compositions, etc., of simple gas mixture and liquid solutions.
	L26-27: Ideal gas mixture model, Fugacity coefficients of pure species, Fugacity coefficients of Species in solution.	T1: 10.4-10.6	
	L28-29: Ideal gas mixture model, Fugacity coefficients of pure species, Fugacity coefficients of Species in solution.	T1: 10.4-10.6	
	T12-T13: Exercise problems on Basic Concepts of Solution Thermodynamics: Theory	T1: Ch10 Problems	
M10: 30 – 34 Solution Thermodynamics: Applications	L30, L31: Liquid-phase properties from VLE data, Activity coefficients from VLE data, Models for Excess Gibbs energy.	T1: 13.1-13.3 10.8-10.9	Apply thermodynamics of simple VLE equilibrium systems in estimation/calculation of changing free energy, enthalpy, entropy, volume, etc., due to mixing in simple systems
	L32: Property changes of mixing.	T1: 11.1	
	L33: Property changes of mixing, Heat effects of mixing processes.	T1: 11.1-11.2	
	L34: Heat effects of mixing processes.	T1: 11.1-11.2	
	T14-T15: Exercise problems on Solution Thermodynamics: Applications.	T1: Ch10 & 11 Problems	
	L35: The reaction coordinate, Application of equilibrium criteria to chemical reactions.	T1: 14.1-14.2	
M11: 35 – 40	L36: The standard Gibbs-Energy changes	T1: 14.3-14.4	1. Apply thermodynamic principles in



Chemical Reaction Equilibria	and equilibrium constant (K), effect of T on K.		estimation/calculation of reaction rate, equilibrium conversion 2. Effect of temperature, pressure, and phase compositions on reaction rate, equilibrium conversion
	L37: Evaluation of K, Relation of K to composition.	T1: 14.5-14.6	
	L38-40: Equilibrium conversion for single reactions, Phase rule & Duham theorem, multireaction equilibrium.	T1: 14.7-14.9	
	T16-T17: Exercise problems on Solution Thermodynamics: Applications.	T1: Ch14 Problems	

7. Evaluation Scheme:

EC No.	Evaluation Component (EC)	Duration	Weightage, % (Marks)	Remarks
1	Tutorial Test*	30 min	20 (60)	CB/OB
2	Mid Semester Test	90 min	30 (90)	CB/OB
3	Comprehensive Exam	180 min	40 (120)	CB/OB
4	Continuous evaluation (Surprise test best 2 out of 4)	During lecture	10 (30)	OB/CB

CB = Close Book **OB** = Open Book

* Each tutorial test will be pre-announced and best 3 tutorial tests out of 4 will be considered for final evaluation.

8. Chamber Consultation Hour: Friday, 5-6 PM

9. Notices: The notices, if any, concerning the course will be communicated on the NALANDA only.

10. Make-up Policy: Make-up will be granted for genuine cases only (No make-up for tutorial tests).

11. NC criterion: 20% of total marks to score grade, else NC will be awarded.

Instructor-in-Charge
 (Course No. CHE F213)



General Handout (Part I):

Review of work, heat, energy for reversible and irreversible processes, First Law applications to closed and open systems, Second law & Entropy, Equations of state and generalized correlations for PVT behaviour, Maxwell relations and fluid properties estimation; Residual and excess properties, Partial molar quantities; Gibbs-Duhem Equation, Fugacity and Activity Coefficient models, Vapour-liquid equilibria, Chemical Reaction Equilibrium.

Prerequisite: NA

Corequisite: NA

SLO Mapping:

S. No.	Description	SLO
CLO 1	Develop ability to analyze heat, energy, and work requirement/transfer for simple physical and chemical processes using laws of thermodynamics.	SLO 1
CLO 2	Applying principles of thermodynamics to chemical engineering systems.	SLO 1
CLO 3	Industrial design of reactors, separators and utilities using principles of thermodynamics.	SLO 2
CLO 4	Thermodynamic efficiency calculation for chemical engineering processes.	SLO 2 & 3
CLO 5	Feasibility of chemical engineering processes based on thermodynamic properties calculations.	SLO 2 & 3